

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #24 — Smart Cities, Transport & Logistics Public Bus Live Tracking & Crowding Estimator

Samarth S Maladkar
SRN: PES1UG24AM396

1. Problem Context & Overview

A municipal transit intelligence platform ingests GPS feeds from city buses, estimates arrival times (ETA) at downstream stops, and computes passenger crowding levels from ticketing sensors. The system supports two primary human stakeholders — the **Commuter**, who wants to know where a bus is and how crowded it will be, and the **Fleet Controller**, who monitors the network and reacts operationally when a route becomes overcrowded or delayed. A **Ticketing Sensor System** supplies the passenger boarding/alighting counts used to derive crowding levels.

Target Stakeholders / Actors: Commuter, Fleet Controller, Ticketing Sensor System (supporting system actor)

2. Requirements Table

Five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001, NFR-002) are defined below. FR-001 and NFR-001 follow the sample requirements provided in the problem statement.

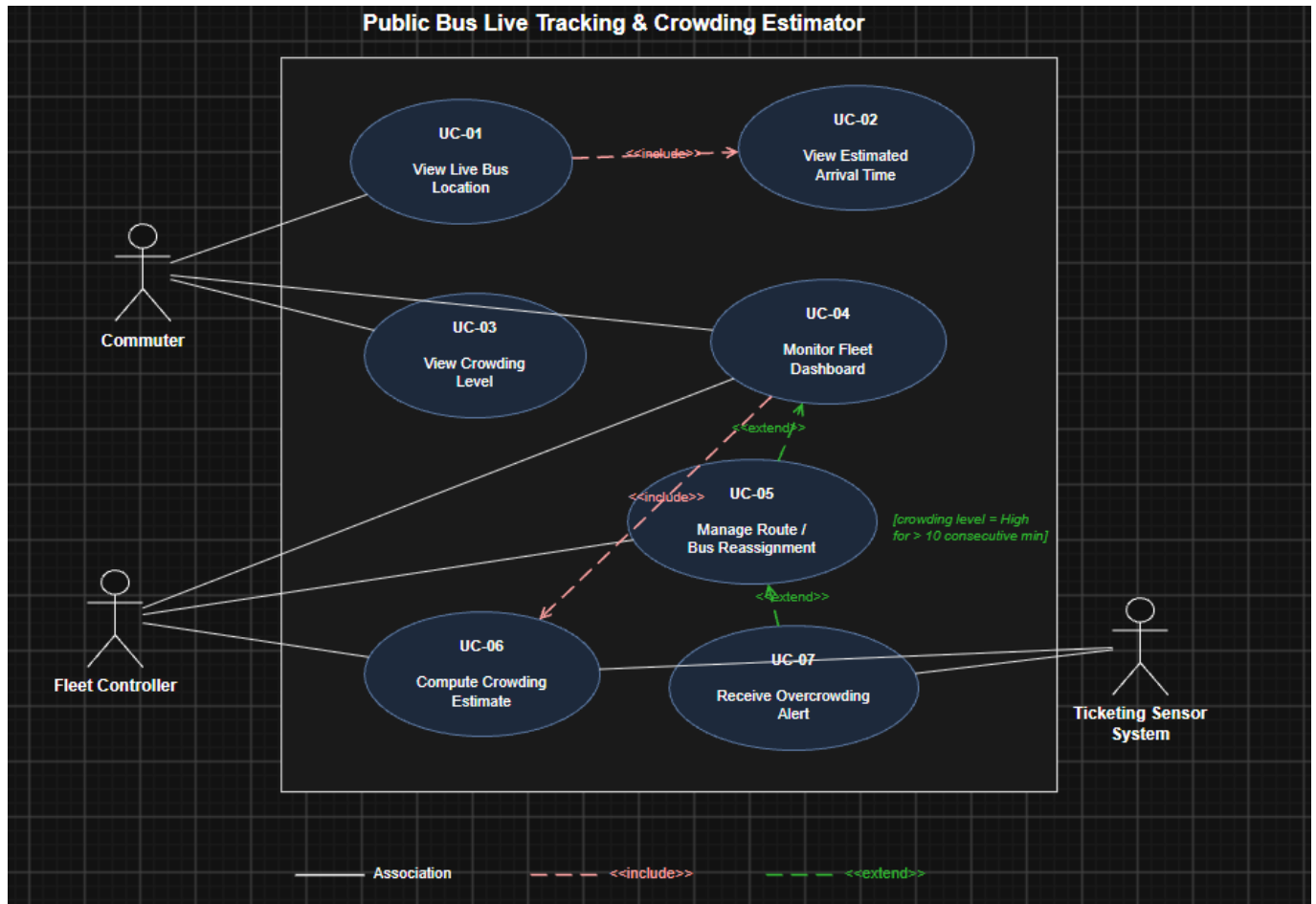
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.	High	Pass: ETA updates based on current traffic speed and distance. Fail: ETA increases while the bus approaches the stop.	Accurate, live ETA is the core value proposition for commuters deciding when to leave for a stop.
FR-002	Functional	The system shall display the live location of every active bus on a selected route to the Commuter on a map view.	High	Pass: the bus marker position updates on the map within 5 seconds of a new GPS ping. Fail: the displayed marker lags the actual GPS timestamp by more than 15 seconds.	Live location visibility builds commuter trust and reduces uncertainty while waiting.
FR-003	Functional	The system shall compute a crowding level (Low / Medium / High) for each active bus using boarding and alighting counts from the ticketing sensors.	Medium	Pass: the computed crowding level matches the expected band within 2% of the calibrated occupancy ratio in test scenarios. Fail: the crowding level remains unchanged despite a significant change in sensor counts.	Lets commuters choose a less crowded bus and gives operations early visibility into demand.
FR-004	Functional	The system shall notify the Fleet Controller when a bus's crowding level remains High for more than 10 consecutive minutes.	High	Pass: an alert appears on the Fleet Controller dashboard within 30 seconds of the threshold being breached. Fail: no alert is raised, or it appears more than 2 minutes late.	Enables timely dispatch of relief buses before overcrowding affects service quality.
FR-005	Functional	The system shall allow the Fleet Controller to reassign an available bus to a route that is overcrowded or delayed.	Medium	Pass: the reassigned bus appears on the target route with a recalculated ETA within 1 minute of confirmation. Fail: the route listing or ETA is not updated after reassignment.	Supports operational responsiveness to real-time demand and delay conditions.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Non-Functional (Performance & Security)	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Pass: benchmarking tests confirm target latency and security standards are met under simulated peak load.	The platform must scale to a city-wide fleet without degrading ETA or crowding accuracy.
NFR-002	Non-Functional (Usability & Reliability)	The commuter-facing app shall remain usable on low-bandwidth (3G) connections and shall continue displaying the last known bus location and ETA during network interruptions of up to 30 seconds.	Medium	Pass: during simulated 3G / intermittent-network tests, the app shows cached data with a 'last updated' timestamp instead of a blank screen. Fail: the app crashes or shows no data during the interruption.	Many commuters rely on mobile data of varying quality; the app must degrade gracefully rather than fail outright.

3. UML Use-Case Diagram

The diagram below models the three actors (Commuter, Fleet Controller, and the Ticketing Sensor System as a supporting system actor), seven use cases (UC-01 to UC-07), and the required <<include>> and <<extend>> relationships:

- <<include>> — **View Estimated Arrival Time** always includes **View Live Bus Location**, since ETA calculation depends on the bus's current position.
- <<include>> — **View Crowding Level** always includes **Compute Crowding Estimate**, since a crowding level cannot be shown without first computing it from sensor data.
- <<extend>> — **Receive Overcrowding Alert** optionally extends **Monitor Fleet Dashboard**, triggered only when a bus's crowding level stays High for more than 10 consecutive minutes.



4. Use-Case Flow Specification

Use Case: UC-02 — View Estimated Arrival Time (ETA)

Primary Actor: Commuter

Supporting Actors: none (system-internal GPS ingestion feed, see FR-001)

Preconditions

- The Commuter has the mobile app open and has selected a route and a stop of interest.
- The system has an active GPS feed (updated at least every 5 seconds, per FR-001) from at least one bus currently servicing the selected route.

Postconditions

- The current ETA for the next arriving bus at the selected stop is displayed to the Commuter.
- The displayed ETA continues to refresh automatically as new GPS pings and traffic data arrive.

Main Success Scenario

1. The Commuter selects a route and a stop from the app's home screen.
2. The system identifies all buses currently active on the selected route (includes UC-01: View Live Bus Location).
3. The system calculates the distance and current travel speed of each identified bus relative to the selected stop, using the most recent GPS ping (per FR-001).
4. The system factors in current traffic conditions along the route to refine the estimate.
5. The system computes an ETA (in minutes) for the next arriving bus and displays it to the Commuter, along with the bus number and current crowding level.
6. The system automatically recalculates and refreshes the ETA every time a new GPS coordinate is ingested (at least every 5 seconds).
7. The Commuter views the live-updating ETA and decides whether to leave for the stop.
8. The use case ends when the Commuter closes the ETA view or the bus arrives at the stop.

Alternate Flow — A1: GPS Signal Lost / Stale Telemetry

- A1a. At step 3, the system detects that the last GPS ping for the tracked bus is older than 30 seconds (signal loss or vehicle GPS fault).
- A1b. The system displays the last known ETA along with a "Signal delayed" indicator and the timestamp of the last known position, instead of a live-updating figure.
- A1c. The system continues attempting to reacquire the GPS feed in the background.
- A1d. If the feed is restored within 60 seconds, the system resumes live ETA updates and the main success scenario continues from step 3.
- A1e. If the feed is not restored within 60 seconds, the system marks the bus as "Tracking unavailable" and suggests the next trackable bus on the route, if any. The use case ends.